

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

Roll No. :52	Name:Roshan Yadav
Class :SYIT	Batch : 0
Date of Assignment :12/08/2026	Date/Time of Submission :12/08/2026

Part – A Theory

5a https://www.w3schools.com/python/pandas/ref_df_filter.asp

5b https://www.w3schools.com/python/module_statistics.asp

5c
<https://www.geeksforgeeks.org/python/creating-a-pandas-series-from-dictionary/>

5d <https://www.geeksforgeeks.org/pandas/boolean-indexing-in-pandas/>

PART – B Practical Questions:

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

Code

```
import pandas as pd
```

```
data = {
    "name": ["Sally", "Mary", "John"],
    "age": [50, 40, 30],
    "qualified": [True, False, False]
}

df = pd.DataFrame(data)
```

```
print(df)
```

Output

```
-- RESUME: C:/USERS/THIO/2
      name  age  qualified
0  Sally   50    True
1  Mary    40   False
2  John    30   False
```

5b. Write a python code to show statistical information on given data set.

Code

import statistics

```
print("mean")
```

```
print(statistics.mean([1, 3, 5, 7, 9, 11, 13]))
```

```
print("\nmedian")
print(statistics.median([1, 3, 5, 7, 9, 11, 13]))
```

```
print("\nmode")
print(statistics.mode([1, 3, 3, 3, 5, 7, 9, 11]))
```

```
print("\nstdev")
print(statistics.stdev([1, 3, 5, 7, 9, 11]))
```

```
print("\nvariance")
print(statistics.variance([1, 3, 5, 7, 9, 11]))
```

Output

```
-----
mean
7

median
7

mode
3

stdev
3.7416573867739413

variance
14
|
```

5c. Write a python code to create pandas series from dictionaries.

Code

```
import pandas as pd
```

```
dictionary = {'A': 10, 'B': 20, 'C': 30}
```

```
series = pd.Series(dictionary)
```

```
print(series)
```

Output

```
-- RESTART: C:/
A      10
B      20
C      30
dtype: int64
|
```

5d. Write a python code to demonstrate filter pandas series with Boolean arrays.

Code

```
import pandas as pd
```

```
dict = {'name':["aparna", "pankaj", "sudhir", "Geeku"],
        'degree': ["MBA", "BCA", "M.Tech", "MBA"],
        'score':[90, 40, 80, 98]}
```

```
df = pd.DataFrame(dict, index = [True, False, True, False])
```

```
print(df)
```

Output

```
== RESTART: C:/Users/info/AppD
   name  degree  score
True  aparna   MBA    90
False  pankaj   BCA    40
True   sudhir  M.Tech   80
False  Geeku   MBA    98
|
```